Multiscale Residual Network With Mixed Depthwise Convolution for Hyperspectral Image Classification

Hongmin Gao, Yao Yang, Chenming Li, Lianru Gao, *Senior Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*— Convolutional neural networks (CNNs) are becoming increasingly popular in modern remote sensing image processing tasks and exhibit outstanding capability for hyperspectral image (HSI) classification. However, for the existing CNN-based HSI-classification methods, most of them only consider single-scale feature extraction, which may neglect some important fine information and cannot guarantee to capture optimal spatial features. Moreover, many state-of-the-art methods have a huge number of network parameters needed to be tuned, which will cause high computational cost. To address the aforementioned two issues, a novel multiscale residual network (MSRN) is proposed for HSI classification. Specifically, the proposed MSRN introduces depthwise separable convolution (DSC) and replaces the ordinary depthwise convolution in DSC with mixed depthwise convolution (MDConv), which mixes up multiple kernel sizes in a single depthwise convolution operation. The DSC with mixed depthwise convolution (MDSConv) can not only explore features at different scales from each feature map but also greatly reduce learnable parameters in the network. In addition, a multiscale residual block (MRB) is designed by replacing the convolutional layer in an ordinary residual block with the MDSConv layer. The MRB is used as the major unit of the proposed MSRN. Furthermore, to enhance further the feature representation ability, the proposed network adds a high-level shortcut connection (HSC) on the cascaded two MRBs to aggregate lower level features and higher level features. Experimental results on three benchmark HSIs demonstrate the superiority of the proposed MSRN method over several state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), high-level shortcut connection (HSC), hyperspectral image (HSI) classification, mixed depthwise convolution (MDConv), multiscale residual block (MRB).**

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I. Introduction

HYPERSPECTRAL images (HSIs) have hundreds of continuous spectral bands with high spectral resolution in the same scene, which can provide rich spectral and spatial information of the ground objects. Currently, HSI has been applied in many fields, including precision agriculture, environmental monitoring, military reconnaissance, and meteorological prediction. Many excellent research works have emerged in HSI analysis and processing fields (e.g., classification [1], change detection [2], endmember extraction [3], and dimensionality reduction [4], [5]). For instance, Du *et al.* [2] proposed a deep slow feature analysis model. It for the first time formulated a deep slow feature space from the time-continuous hyperspectral data, which proves an enlightening approach to suppress the unchanged components and highlight the changed components for the hyperspectral remote sensing images. HSI classification is the first and most important step to realize many hyperspectral remote sensing applications. The classification effect will directly affect the accuracy of information acquisition from the HSIs and further affect the subsequent application effect. The traditional HSI-supervised classification methods usually develop the feature-engineering approaches [6], [7] to design features manually and then classify these features with the vanilla classifier, such as support vector machine [8], neural network [9], and logistic regression [10]. In [11]–[14], many spectral–spatial feature-based classification methods (e.g., extended morphological profiles [11] and composite kernel [12]) have been developed. Although many methods have obtained good performance, they are equipped with a limited representation capacity and can only extract low-level features due to the shallow nonlinear structure. Moreover, the intrinsic complexity and rather limited labeled samples of HSI often make it difficult for them to provide satisfactory classification accuracies.

In recent years, many deep-learning-based HSI-classification methods have been proposed and showed excellent performance. Compared with the traditional handcrafted feature-based classification methods, deep-learning-based methods are able to extract automatically the high-level abstract features from the HSIs in a hierarchical learning way. In addition, they have more powerful feature-representation ability due to their deep nonlinear network

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architectures. At first, the stacked autoencoder (SAE) [15] and the deep belief network (DBN) [16] were introduced for the feature extraction and classification of the HSIs. In the following research works, some improved SAE-based methods [17], [18] and DBN-based methods [19] have been proposed to achieve better classification results. Although these methods can effectively extract the deep features, they require flattening the input image patches to the 1-D vectors before fed to the networks, which will destroy the spatial context correlation and cause the loss of useful spatial information. To address this issue, many 2-D convolutional neural network (CNN) models were proposed for HSI classification [20]–[25], which can directly process the 3-D image patches and extract the spatial context information more effectively. For example, the CNN was combined with the pixel pairs to learn the discriminative features in [22], and a majority of voting strategy was adopted to obtain the final classification results. Cao *et al.* [24] proposed a Bayesian HSI-classification method, where a CNN was used to extract the spectral–spatial features from the 3-D patches, and a smooth Markov random field prior was placed on the labels to exploit further the spatial information. Ma *et al.* [25] introduced the unpooling and deconvolution operations, and proposed a deconvolution network with skip connection for HSI classification. Furthermore, Li *et al.* [26] and Hamida *et al.* [27] designed 3-D CNNs for HSI classification, which can simultaneously process the HSI information in spatial and spectral dimensions. In [28], a recurrent 3-D CNN model was advocated to classify the HSIs, which can converge fast for its recurrent network structure.

Although the CNN-based methods have achieved promising progress in HSI classification, they need a large number of labeled samples to train many learnable parameters in the network, while the available labeled samples are rather limited in the HSI, which causes the so-called small sample size (3S) problem. In general, a deeper network can learn a more abstract feature representation from the input data. However, it also means there are more training parameters needed to be tuned and the overfitting problem caused by the 3S problem will be more serious, thus bringing about poor classification performance. In addition, the gradient vanishing is likely to appear when training the network. To deal with the above issues, some works introduced residual connection [29] into their deep networks, so as to alleviate overfitting and enhance the generalization performance. For instance, Song *et al.* [30] proposed an HSI-classification method based on the deep feature-fusion network (DFFN), which can consider the correlated information among different layers by a very deep architecture with residual connection and fuse mechanism. Zhong *et al.* [31] designed a spectral–spatial residual network (SSRN) with two consecutive residual blocks to learn the discriminative features for the HSIs, which can provide excellent performance in the case of small training sample set. In [32], a deep pyramidal residual network was developed to learn more robust spectral–spatial representations from the HSI cubes. Moreover, Wang *et al.* [33] and Fang *et al.* [34] adopted dense connection to design deep networks for HSI classification. Those methods achieved the state-of-the-art classification results. However, due to the very deep architecture or the use of the 3-D convolutional layers in the CNN, there are a huge number of training parameters in their networks, which causes expensive computational cost and slows down the classification process. In addition, they can only extract single-scale features from the fixed-size image patches, which may neglect some important fine information and cannot guarantee to capture the optimal spatial context information.

More recently, some works developed the CNN models to extract the multiscale spectral–spatial features for HSI classification. Lee and Kwon [35] designed a contextual deep CNN (CD-CNN), where a multiscale filter bank was placed on the first layer of the network to extract the multiscale features from the single-scale input patches. Similarly, Gong *et al.* [36] developed three types of CNN structures with the multiscale CNNs (MS-CNNs) for 1-D spectral, and 2-D and 3-D spectral–spatial HSI classifications, respectively. In addition, a diversified metric was combined with the MS-CNNs to obtain better performance. In [37], a pretrained VGG-16 was applied to extract the multiscale spatial structures and an unsupervised cooperative sparse autoencoder method was proposed to fuse the deep spatial features and spectral information. Unfortunately, the works [35] and [36] only perform multiscale feature extraction at a certain layer of the network, which may not be enough for finely and comprehensively learning the spectral and spatial context informations at different scales. The work [37] does not take full advantage of the joint spectral–spatial information, and the implementation process of the method is slightly complicated. To extract more discriminative features, Li *et al.* [38] proposed a multiscale deep middle-level feature-fusion network (MMFN) to learn different scale features from the multiscale input image patches. The MMFN first trained a CNN model for each scale input separately and saved the optimal models. The middle-level features were obtained by the corresponding scale model, which can guarantee to obtain the optimal multiscale middle-level features. Then, the MMFN adopted a convolutional layer to fuse those multiscale middle-level features and trained a new CNN model for final classification. In [39], a hierarchical multiscale CNN with the auxiliary classifier (HMCNN-AC) framework was proposed. In the HMCNN-AC, multiscale CNNs were designed to extract the spectral–spatial features at different input scales simultaneously, and a bidirectional long-short-term-memory (LSTM) network was adopted to capture the dependence and the correlation between the obtained multiscale features. Moreover, weighted auxiliary classifiers are employed to train the whole network in a better way. Zhao and Du [40] also proposed a multiscale CNN to capture the contextual information at different scales for HSI classification. Although those methods [38]–[40] using multiscale inputs have achieved remarkable performance, they not only cause spatial information redundancy but also cause extra computation cost and even make it to get multiplied, thus significantly increasing the time consumption of the classification tasks.

Considering the limitations of those methods based on multiscale feature extraction [35]–[40], this article proposes

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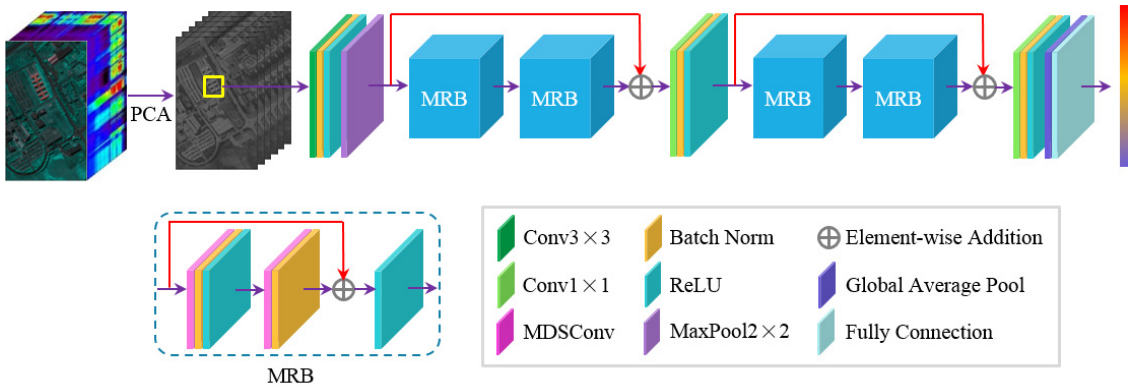


Fig. 1. Framework of the proposed MSRN-based HSI-classification method.

a novel multiscale residual network (MSRN) for HSI classification. The mixed depthwise convolution (MDConv) studied in [41] is adopted in our proposed MSRN, which can extract the spatial features of different scales by mixing up different kernel sizes in a single convolution operation. The main contributions of this article can be summarized as follows.

First, to reduce the training parameters and speed up the classification process, we adopt the depthwise separable convolution (DSC) [42] to design a lightweight CNN for HSI classification. To the best of our knowledge, it is the first time to apply the DSC in HSI classification.

Second, to achieve multiscale feature learning, a multiscale residual block (MRB) is proposed. Specifically, we replace the ordinary depthwise convolution in the DSC with the mixed depthwise convolution, thus developing a DSC with mixed depthwise convolution (MDSConv). In addition, an MRB is designed by incorporating the MDSConv into a residual block. The residual learning can alleviate the overfitting effectively. The MRB is able to perform lightweight and efficient multiscale feature extraction.

Finally, to extract the discriminative features more robustly, we add a high-level shortcut connection (HSC) on two cascaded MRBs, which aggregates the lower level features and the higher level features by summation. The proposed MSRN is able to carry out multiscale feature extraction at different levels of the network, thus allowing more careful and comprehensive feature representation for the HSI.

The remainder of this article is organized as follows. Section II gives a detailed description of our proposed HSI-classification method. Section III reports the experimental results and evaluates the performance of the proposed method. Finally, Section IV presents the final conclusion and discusses future research works.

II. Methodology

Some previous studies [35]–[40] have demonstrated that taking full advantage of the different-scale information is helpful for HSI classification. However, those methods using single-scale inputs usually only execute multiscale feature extraction once, which is hard to learn comprehensively the spectral–spatial feature representations at different scales, while those methods using multiscale inputs always cause extra computation cost and longer training time. Furthermore, many state-of-the-art HSI-classification methods have a great number of training parameters, which will cause high computation cost. To address the abovementioned two issues, we design a novel MSRN model for HSI classification, which is able to carry out multiscale feature extraction at different levels of the network. Fig. 1 shows the framework of the proposed classification methods. There exist some redundant bands, and they can only provide very little useful information, which means they make little contribution on the classification task. They may even cause a negative impact on the final classification because of the noise existing in them. Therefore, first, the principal component analysis (PCA) is adopted to remove some noise and those unimportant dimensions from the original HSI. In this way, most of the useful information is retained. With that retained information, CNNs are able to extract enough discriminative features for the correct classification of the ground objects. Moreover, due to that many dimensions are removed, PCA reduces the computation cost of the deep network to a certain extent. Then, the image patches centered on the labeled pixels are obtained and fed into the MSRN, where some patches are used for training the network and the rest of the patches are used for evaluating the performance of the trained network. Finally, the labels of test pixels can be predicted by the trained MSRN. As can be seen from Fig. 1, the MRB is the main part of our proposed MSRN. In addition, the MDSConv layer is the main part in the MRB, which can greatly reduce the parameter number of the network. The architectures of MDSConv and MRB are detailed as follows.

Mixed Depthwise Convolution DSC is attracting increasing attention to the modern CNN architecture design studies. Because it can not only greatly reduce the number of network parameters, but also can almost not lose the accuracy, or even increase the accuracy. Fig. 2 gives the diagram of ordinary convolution, DSC, and MDSConv operation. As we can see, the DSC divides the ordinary convolution into two parts. The one is the depthwise convolution and the other is pointwise convolution, which is the 1×1 convolution in essence. The former convolves each channel of the input feature map separately, thus capturing the spatial features of each channel. The latter integrates all extracted spatial features and learns the

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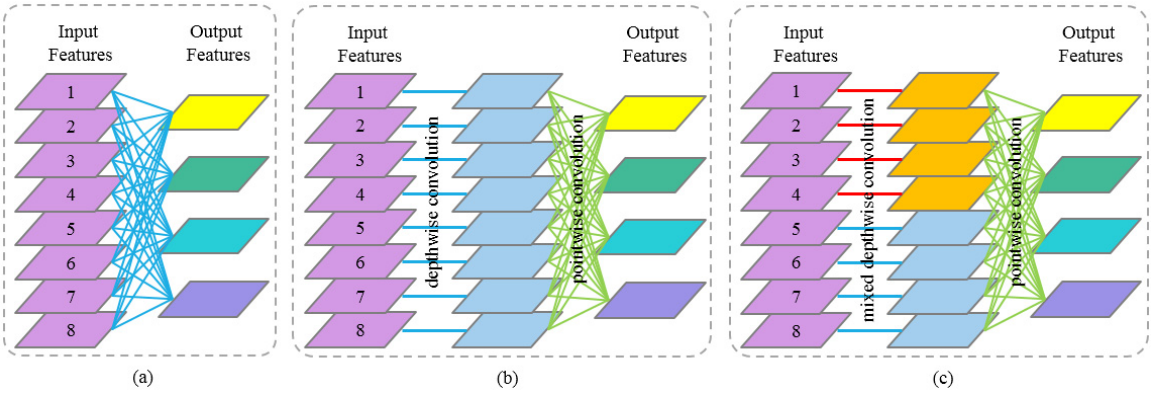


Fig. 2. Three kinds of convolution operation. (a) Ordinary convolution. (b) DSC. (c) Proposed MDSConv. Each parallelogram means one certain channel of the feature map and the connection with different colors means the convolution filter with different sizes.

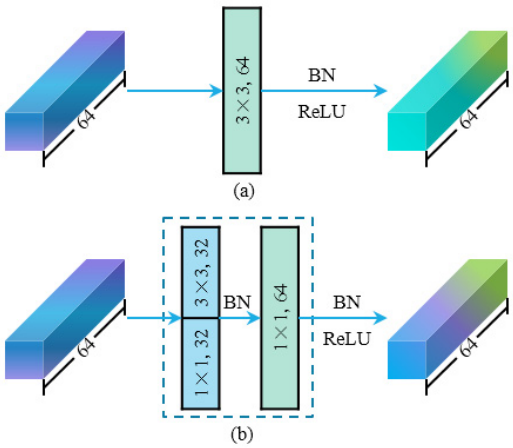


Fig. 3. Example for comparing the parameter number between (a) ordinary 3×3 convolutional layer and (b) MDSConv layer.

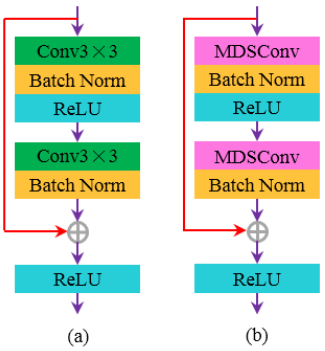


Fig. 4. Architectures of (a) ordinary residual block and (b) proposed MRB.

channel correlation information of the input feature map. The ordinary 2-D convolution operation considers all the channels of the corresponding region simultaneously. In order to achieve multiscale feature extraction, we mix up the 1×1 filter and the 3×3 filter into a depthwise convolution operation, and the number of two types of filters is equal, which is half of the channel number of the input feature map. In other words, for each feature map, half of its channels are convolved with the 1×1 filter and the rest of the channels are convolved with the 3×3 filter. As shown in Fig. 2(c), the MDSConv is developed by incorporating the MDConv into the DSC. In this way, the spatial correction can be learned at different scales.

Let C_{in} and C_{out} denote the number of input channels and output channels of an ordinary 3×3 convolutional layer, respectively. Without considering the bias, the parameter number P_1 of this layer is $C_{\text{in}} \times C_{\text{out}} \times 3 \times 3$. Assuming that a DSC layer with 3×3 depthwise convolution has the same input and output channel number as the ordinary 3×3 convolutional layer. Then, the number of parameters P_2 in this layer can be calculated as $C_{\text{in}} \times 3 \times 3 + C_{\text{in}} \times C_{\text{out}}$. In addition, the parameter number P_3 of a proposed MDSConv layer can be calculated as $C_{\text{in}}/2 \times 3 \times 3 + C_{\text{in}}/2 \times 1 \times 1 + C_{\text{in}} \times C_{\text{out}}$. For example, as shown in Fig. 3, we assume that both C_{in} and C_{out} are 64, and then, P_1, P_2, and P_3 are 36 864, 4672, and 4416, respectively. Therefore, the parameters of an MDSConv layer are less than 1/8 of that of an ordinary 3×3 convolutional layer. It means the proposed MDSConv can speed up the training and test processes. In addition, the MDSConv allows the proposed MSRN learn more discriminative spectral–spatial feature representation for better accuracy.

Multiscale Residual Block The residual learning is introduced into our study to construct the deep hierarchical architecture with good generalization performance. Fig. 4(a) gives the architecture of the ordinary residual block. As can be observed, the ordinary residual block contains two 3×3 convolutional layers and a shortcut connection (also known as skip connection). In addition, the batch normalization (Batch Norm) is usually applied after the convolutional layer and before activation function, so as to accelerate the convergence of the network. Each residual block can be expressed as

$$F(X) = f(G(X) + H(X)) \tag{1}$$

where X and $F(X)$ denote the input and output of the residual block, respectively, and $f(\cdot)$ denotes the activation function. In addition, $G(\cdot)$ denotes a residual learning function, which performs a nonlinear transformation with a series of operations including convolution, Batch Norm, and rectified linear units (ReLUs). Therefore, $G(X)$ denotes the output of the Batch Norm before the summation operation. The $H(X) = X$ is an identity mapping function, which is achieved by using

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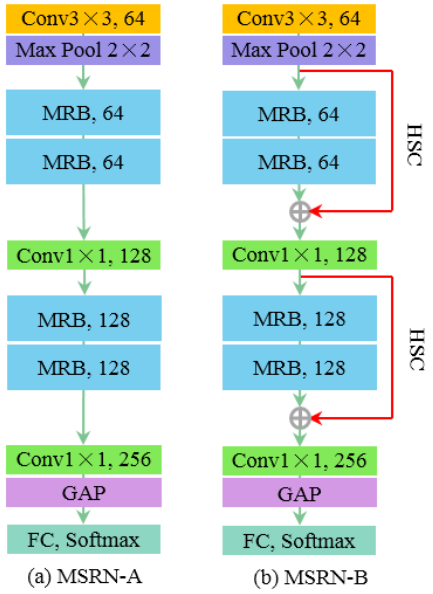


Fig. 5. Architectures of the proposed two networks. (a) MSRN-A. (b) MSRN-B. For convenience, the batch normalization, activation layers immediately after the convolutional layer are not provided. The number "64/128/256" refers to the filter number of the corresponding convolutional layers.

the shortcut connection. The shortcut connection enables the gradient propagate directly from the later to earlier layers, thus mitigating the gradient vanishing that usually exists in the deep network. By stacking multiple residual blocks, one can develop a deeper network and effectively alleviate overfitting of the network.

As shown in Fig. 4(b), this article designs the MRB by replacing the convolutional layers in the ordinary residual block with the MDSConv. The MRB is capable of extracting multiscale spatial features and spectral–spatial correlated information, and it has much fewer parameters than the ordinary residual block. Therefore, by staking multiple MRBs, our proposed MSRN can perform multiscale feature extraction at different levels of network and learn more careful and comprehensive spectral–spatial feature representations of the HSIs for classification. In addition, the network will converge faster and spend less time for HSI classification.

A. Multiscale Residual Networks

Based on the MRB, this article designs two networks, including MSRN-A and MSRN-B. As shown in Fig. 5, the difference between these two networks is that the MSRN-B adds extra HSC on the two cascaded MRBs without increasing any parameters. By using the HSC, the MSRN-B can aggregate the lower level features and the higher level multiscale features by summation. For all convolutional layers in both networks, the stride is 1 and the padding is used, which means the spatial size of each feature map will remain unchanged after fed into any convolutional layer. The number of filters in each convolutional layer is given in Fig. 5. Here, we take the Pavia University (PU) data set as an example to explain the implementation details of the designed two networks. First, the 3-D image patches with the size of $15 \times 15 \times 10$ are fed into an ordinary 3×3 convolutional layer that has 64 filters, and the $15 \times 15 \times 64$ feature map is generated. Then, the generated feature maps are subsampled by a max pooling layer with a 2×2 kernel and stride 2, so as to reduce the computation cost of the network. The size of each feature map becomes $8 \times 8 \times 64$. Next, two MRBs are staked to extract the hierarchical multiscale features. After that, an ordinary 1×1 convolutional layer increases the channels of each feature map from 64 to 128, which makes multiscale features more abundantly and fully represented. Then, feature maps with the size of $8 \times 8 \times 128$ are transmitted to the two cascaded MRBs where each convolutional layer has 128 filters. Again, another ordinary 1×1 convolutional layer is employed to enhance feature representation and change the channels of the feature map from 128 to 256. Finally, a global average pooling (GAP) layer is adopted to extract the discriminative features for final classification, which transforms the $8 \times 8 \times 256$ feature map to a $1 \times 1 \times 256$ feature tensor. The fully connection (FC) layer can adaptively generate a $1 \times L$ feature vector, where L is equal to the number of land-cover classes in the HSI. Different from MSRN-A, in the MSRN-B, the multiscale features extracted by the second MRB are not directly passed on to the next layer, but aggregated with the output features of the pooling layer. Similarly, the features extracted by the fourth MRB are first aggregated with the input features of the third MRB and then processed by the following layer.

III. Experimental Results and Analysis

A. Data Set Description

Three HSI data sets are adopted to evaluate the performance of the proposed method, including the Salinas (SA) image, the PU image, and the Pavia Center (PC) image.

The SA image was captured by the AVIRIS sensor over the SA Valley in California. It consists of 224 spectral bands and has 512×217 pixels with the spatial resolution of 3.7 m. There are 20 water absorption bands removed and only 204 bands are reserved for the experiment. It contains 16 land-cover classes and 54 129 labeled pixels.

The PU image was captured by the ROSIS sensor over PU in northern Italy. There are 115 spectral bands ranging from 430 to 860 nm and 610×340 pixels with the spatial resolution of 1.3 m. After removing the noise-affected bands, 103 bands remained for the experiment. In addition, it contains nine land-cover classes and 42 776 labeled pixels.

The PC image was captured by the ROSIS sensor over Pavia City in northern Italy. It has 1096×1096 pixels with the spatial resolution of 1.3 m and 115 spectral bands covering from 400 to 1000 nm. After removing a black region of 1096×381 pixels and 13 noise-affected bands, the remaining region becomes a two-part image with 102 spectral bands, and the right part of 1096×492 pixels is used for our experiments. There are nine land-cover classes and 103 539 labeled pixels within this image.

For each data set, 20 samples per class are randomly selected (from all the labeled samples) as training samples, and the rest of the labeled samples are used for testing. Tables I–III report the number of training samples and test

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES IN THE SA DATA SET

No.	Name	Number	Train	Test
1	green_weeds_1	2009	20	1989
2	green_weeds_2	3726	20	3706
3	Fallow	1976	20	1956
4	Fallow_rough_plow	1394	20	1374
5	Fallow_smooth	2678	20	2658
6	Stubble	3959	20	3939
7	Celery	3579	20	3559
8	Grapes_untrained	11271	20	11251
9	Soil_vinyard	6203	20	6183
10	Corn_senesced	3278	20	3258
11	Lettuce_romaine_4wk	1068	20	1048
12	Lettuce_romaine_5wk	1927	20	1907
13	Lettuce_romaine_6wk	916	20	896
14	Lettuce_romaine_7wk	1070	20	1050
15	Vinyard_untrained	7268	20	7248
16	Vinyard_vertical	1807	20	1787
	Total	54129	320	53809

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES IN THE PU DATA SET

No.	Name	Number	Train	Test
1	Asphalt	6631	20	6611
2	Meadows	18649	20	18629
3	Gravel	2099	20	2079
4	Trees	3064	20	3044
5	Painted metal sheets	1345	20	1325
6	Bare Soil	5029	20	5009
7	Bitumen	1330	20	1310
8	Self-Blocking Bricks	3682	20	3662
9	Shadows	947	20	927
	Total	42776	180	42596

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES IN THE PC DATA SET

No.	Name	Number	Train	Test
1	Water	65278	20	65258
2	Trees	6508	20	6488
3	Meadows	2905	20	2885
4	Self-Blocking Bricks	2140	20	2120
5	Bare soil	6549	20	6529
6	Asphalt	7585	20	7565
7	Bitumen	7287	20	7267
8	Tiles	3122	20	3102
9	Shadows	2165	20	2145
	Total	103539	180	103359

TABLE IV
OA (%) OBTAINED BY MSRN-B WITH DIFFERENT NUMBER OF PRINCIPAL COMPONENTS

N	SA	PU	PC
5	94.70±0.52	92.64±0.63	98.06±0.13
10	96.30±0.40	96.50±0.47	98.69±0.12
15	96.84±0.18	95.81±0.08	98.46±0.11
20	96.65±0.21	95.46±0.85	98.23±0.07

TABLE V
OA (%) OBTAINED BY MSRN-B WITH DIFFERENT PATCH SIZES

Patch size	SA	PU	PC
5×5	——	——	97.71±0.17
7×7	——	——	98.09±0.28
9×9	94.54±0.52	94.41±0.59	98.69±0.12
11×11	95.53±0.34	95.45±0.30	98.12±0.10
13×13	96.24±0.53	96.18±0.53	97.85±0.31
15×15	96.84±0.18	96.50±0.47	——
17×17	97.12±0.43	96.78±0.37	——

samples for all data sets. Note that all input data of three data sets are standardized to zero mean and unit variance before fed into the network.

B. Experimental Setup

The overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are adopted to evaluate quantitatively the classification performance. Specifically, the OA refers to the ratio of the total number of the correctly classified samples to that of all test samples. The AA is the average of the classification accuracy of all classes. The κ is used to assess the agreement of classification for all the classes. For each experiment, we run ten times with the randomly selected training samples and report the mean and standard deviation for all the metrics. The He-initialization method [43] is adopted to initialize the weight parameters of the proposed networks. The Adam optimizer is adapted to optimize the learnable parameters. For all three data sets, the learning rate and batch size are 0.01 and 64, respectively. The total epochs are set as 120, 120, and 60 for the SA, PU, and PC data sets, respectively. All experiments are conducted with TensorFlow 1.4 on a desktop PC with Intel Core i5-4460 CPU and 8.00-GB RAM. The experiment mainly contains three parts. First, with 20 training samples per class, we analyze the effect of some hyperparameters on the performance of the proposed MSRN-B and discuss the parameter setting for three data sets. Second, the classification performance and computing time of the proposed method are compared with several related methods on three data sets. Third, we adjust the arrangement of MRB in MSRN-B, so as to observe the effect of location of performing multiscale feature extraction on the classification of an HSI.

C. Parameters Analysis

The number of principle components and spatial size of the image patches are two important hyperparameters that

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TABLE VI
CLASSIFICATION RESULTS OF THE SA DATA SET OBTAINED BY DIFFERENT METHODS WITH 20 TRAINING SAMPLES PER CLASS

No.	3D-CNN	DCNN	DRN	CD-CNN	DFFN	SSRN	MSRN-A	MSRN-B
1	99.66±0.31	99.77±0.35	99.78±0.48	98.95±0.61	99.97±0.04	99.90±0.12	99.95±0.12	100.00±0.00
2	99.29±0.46	100.00±0.00	99.85±0.33	98.93±0.41	99.89±0.17	99.84±0.10	99.01±0.62	99.44±0.65
3	96.44±1.43	99.30±1.06	99.31±1.36	92.98±1.40	99.91±0.13	99.82±0.20	99.97±0.06	100.00±0.00
4	92.26±10.18	94.57±2.28	94.16±1.83	96.26±1.89	84.10±3.21	96.95±0.87	96.95±0.67	95.88±1.06
5	96.67±1.67	94.34±3.41	98.60±0.45	96.97±3.20	99.26±0.44	99.29±0.40	98.75±0.53	98.94±0.32
6	99.88±0.12	98.06±1.01	99.71±0.31	98.22±1.20	99.98±0.02	99.99±0.01	99.95±0.09	100.00±0.00
7	98.27±1.83	99.98±0.05	99.60±0.50	98.24±0.82	99.94±0.08	99.58±0.23	99.93±0.14	99.81±0.50
8	76.43±3.46	94.36±2.05	95.93±2.22	80.46±3.00	97.87±1.51	91.01±1.21	96.00±0.89	97.36±0.75
9	98.28±0.67	99.66±0.50	100.00±0.00	99.69±0.19	100.00±0.00	99.58±0.01	99.99±0.01	99.99±0.01
10	85.42±3.83	94.51±2.22	98.80±1.23	86.60±2.15	99.95±0.05	91.79±1.36	99.75±0.33	99.44±0.79
11	81.81±14.49	98.27±1.42	94.87±7.33	81.52±3.04	97.67±1.00	97.17±1.54	95.89±2.49	96.04±1.78
12	96.13±2.34	96.77±7.19	99.95±0.07	98.21±1.93	100.00±0.00	99.84±0.10	100.00±0.00	99.99±0.02
13	92.36±7.57	98.59±1.34	98.74±1.66	83.06±5.92	98.28±0.84	99.70±0.39	99.61±0.44	99.58±0.18
14	88.55±5.41	94.55±2.86	95.09±3.13	82.35±5.11	84.15±4.85	94.23±1.28	93.11±3.84	94.08±1.79
15	56.37±3.58	79.04±0.87	80.14±3.69	61.77±3.58	86.50±1.41	78.32±3.44	83.42±1.09	85.50±0.67
16	97.71±1.23	99.95±0.10	100.00±0.00	95.49±1.30	99.82±0.31	99.15±0.83	100.00±0.00	100.00±0.00
OA (%)	85.71±0.93	94.37±0.37	95.40±0.41	87.17±0.77	96.56±0.19	94.08±0.79	96.23±0.14	96.84±0.18
AA (%)	90.97±0.99	96.36±0.58	97.16±0.49	90.61±0.62	96.71±0.38	96.64±0.34	97.64±0.27	97.88±0.16
$\kappa\times100$	84.15±1.03	93.75±0.40	94.89±0.45	85.76±0.84	96.17±0.21	93.42±0.87	95.80±0.15	96.48±0.20

TABLE VII
CLASSIFICATION RESULTS OF THE PU DATA SET OBTAINED BY DIFFERENT METHODS WITH 20 TRAINING SAMPLES PER CLASS

No.	3D-CNN	DCNN	DRN	CD-CNN	DFFN	SSRN	MSRN-A	MSRN-B
1	92.41±1.50	95.64±0.75	96.76±1.21	92.75±0.38	89.59±3.33	97.53±0.92	96.95±1.09	97.51±0.74
2	90.86±2.63	97.35±1.13	98.31±1.09	93.79±1.40	96.58±0.53	98.93±0.23	98.93±0.55	99.60±0.12
3	60.67±10.84	90.73±2.93	91.71±4.89	51.54±6.77	88.32±5.80	90.76±4.48	95.61±4.60	95.10±2.55
4	81.55±6.98	85.90±4.42	78.33±11.09	94.45±1.06	63.72±5.44	91.31±7.23	90.50±6.24	88.55±1.21
5	96.03±7.51	97.53±1.35	96.24±1.28	96.03±1.54	86.03±1.36	99.03±0.67	93.66±2.24	93.53±2.43
6	50.58±5.84	92.27±6.32	95.73±3.25	67.36±4.30	89.25±4.93	73.65±5.88	97.14±1.95	92.77±3.05
7	49.44±4.40	87.18±13.25	86.67±7.15	56.51±5.18	72.21±11.07	99.24±0.52	90.86±6.72	99.30±0.64
8	72.05±5.33	74.50±8.54	81.31±4.68	69.77±3.99	79.38±5.29	91.55±6.40	83.03±5.39	92.95±2.32
9	91.70±9.40	92.10±3.68	92.19±3.69	97.64±1.53	74.93±6.20	99.94±0.11	90.94±3.32	97.07±0.36
OA (%)	77.85±2.65	92.18±0.66	93.20±1.00	83.70±1.24	87.64±0.70	92.96±0.98	95.37±1.14	96.50±0.47
AA (%)	76.14±3.07	90.36±1.63	90.80±1.45	79.98±1.01	82.22±1.44	93.55±0.78	93.07±1.35	95.15±0.62
$\kappa\times100$	71.56±3.27	89.69±0.86	91.09±1.26	78.68±1.57	83.83±0.88	90.86±1.24	93.89±1.48	95.38±0.62

can control the performance of the proposed method. In this section, these two hyperparameters are analyzed by using three data sets.

Table IV reports the classification results obtained by MSRN-B on three data sets under different numbers (N) of principle components. It can be observed that as the N increases, the OA first increases and then slightly decreases for each data set. Less principle components will cause spectral information cannot be fully used, thus decreasing the classification accuracy. In addition, too much principle components will cause redundant information and higher computing cost. Therefore, the N is set to be 15, 10, and 10 for the SA, PU, and PC data sets, respectively.

Table V shows the effect of patch size on the performance (OA) of MSRN-B. As can be seen, as the patch size increases, the OA continuously increases for the SA and PU data sets. However, enlarging the patch size just minorly improves the OA when it reaches 15×15. For the PC data set, the OA first increases and then decreases. Considering that a larger patch size will cause higher computational cost, the patch size is set to be 15×15, 15×15, and 9×9 for the SA, PU, and PC data sets, respectively.

D. Comparison Results of Different Methods

The performance of the proposed MSRN is compared with that of a deep residual network (DRN) [29], a deep

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TABLE VIII
CLASSIFICATION RESULTS OF THE PC DATA SET OBTAINED BY DIFFERENT METHODS WITH 20 TRAINING SAMPLES PER CLASS

No.	3D-CNN	DCNN	DRN	CD-CNN	DFFN	SSRN	MSRN-A	MSRN-B
1	99.98±0.02	99.99±0.01	99.99±0.01	99.97±0.03	99.97±0.03	100.00±0.00	100.00±0.00	100.00±0.00
2	96.91±1.61	96.29±1.21	99.53±0.25	95.79±0.41	97.88±1.05	99.02±0.20	98.92±0.93	99.47±0.15
3	89.64±3.29	89.60±3.77	91.47±1.30	88.29±1.72	84.58±4.13	94.51±0.59	93.05±1.29	89.64±3.11
4	61.34±1.16	60.12±1.05	62.68±1.01	58.93±1.46	61.97±7.34	62.68±0.50	67.89±4.08	76.31±5.04
5	99.25±0.24	99.01±0.33	99.15±0.14	98.98±0.24	94.86±2.99	99.82±0.21	99.31±0.37	99.34±0.30
6	91.39±1.47	91.83±3.50	96.85±0.98	89.51±0.92	92.47±2.80	94.64±0.59	95.67±1.48	96.98±0.25
7	88.80±3.34	89.23±3.63	88.17±7.39	89.32±3.11	91.38±10.35	92.02±1.34	95.81±4.90	99.93±0.12
8	99.63±0.39	99.51±0.21	99.65±0.18	99.13±0.98	97.45±2.15	99.68±0.46	99.59±0.24	99.73±0.17
9	99.98±0.04	99.65±0.33	97.67±3.16	99.58±0.50	84.97±7.28	100.00±0.00	98.93±0.45	98.58±0.61
OA (%)	96.75±0.26	96.68±0.24	97.27±0.62	96.38±0.23	96.15±1.18	97.61±0.09	98.07±0.23	98.69±0.12
AA (%)	91.88±0.54	91.69±0.61	92.79±0.84	91.06±0.39	89.50±2.24	93.60±0.14	94.35±0.18	95.55±0.41
$\kappa\times100$	94.43±0.45	94.31±0.42	95.36±1.02	93.80±0.39	93.45±1.94	95.91±0.15	96.68±0.38	97.75±0.21

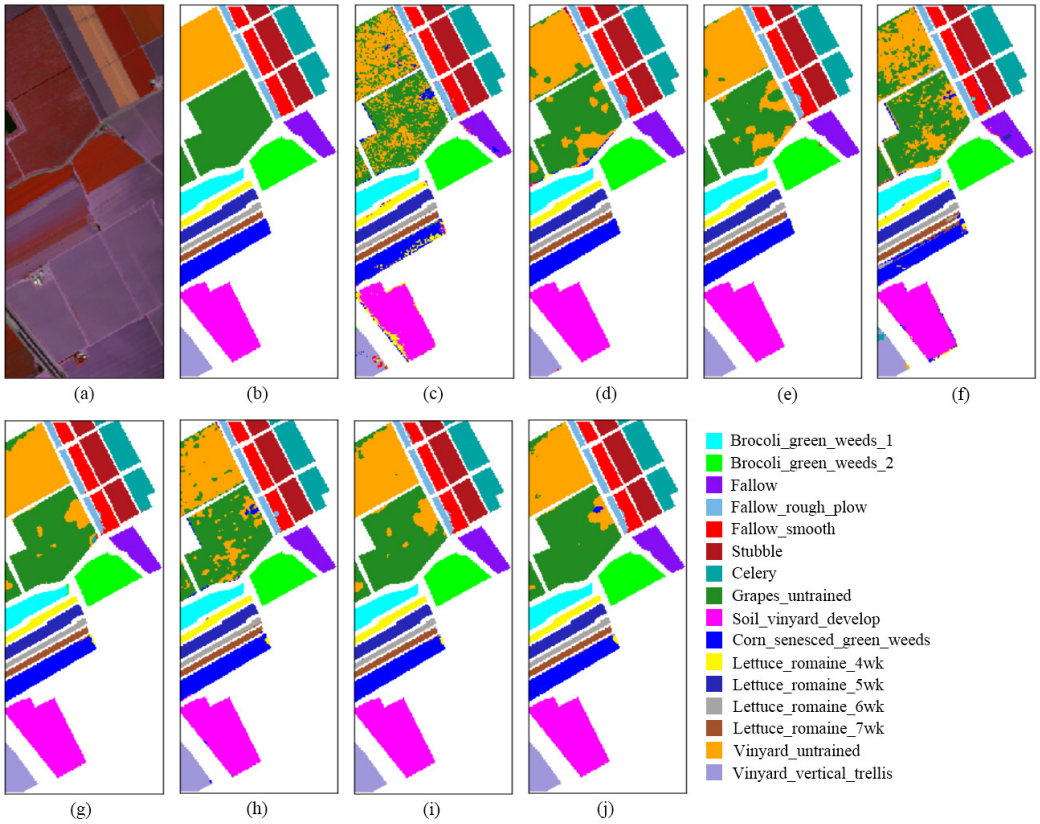


Fig. 6. Classification maps of the SA data set obtained by different methods with 20 training samples per class. (a) False-color composite image. (b) Ground truth. (c) 3-D-CNN (OA = 85.68%). (d) DCNN (OA = 94.29%). (e) DRN (OA = 95.52%). (f) CD-CNN (OA = 87.08%). (g) DFFN (OA = 96.56%). (h) SSRN (OA = 94.10%). (i) MSRN-A (OA = 96.29%). (j) MSRN-B (OA = 96.86%).

CNN (DCNN), and four state-of-the-art methods including 3-D-CNN [26], CD-CNN [35], DFFN [30], and SSRN [31]. Specifically, all compared methods adopt 3-D image patches as the inputs of the network. Note that the DRN, DCNN, and MSRN-A have the similar architecture. The DRN can be obtained by replacing all MRBs in the MSRN-A with the ordinary residual blocks. In other word, the MSRN-A is provided with MDSConv, but the DRN is not. The DCNN is a plain network without using residual learning, and it can be obtained by removing all the shortcut connections of the DRN. The DRN and DCNN have the same input patches as the proposed MSRN. The 3-D-CNN is composed of two 3-D convolutional layers and an FC layer. The other three methods, CD-CNN, DFFN, and SSRN, all adopt residual connection to design their deep architectures. More specifically, there are more than 20 or even 30 weight layers in DFFN for different data sets. The

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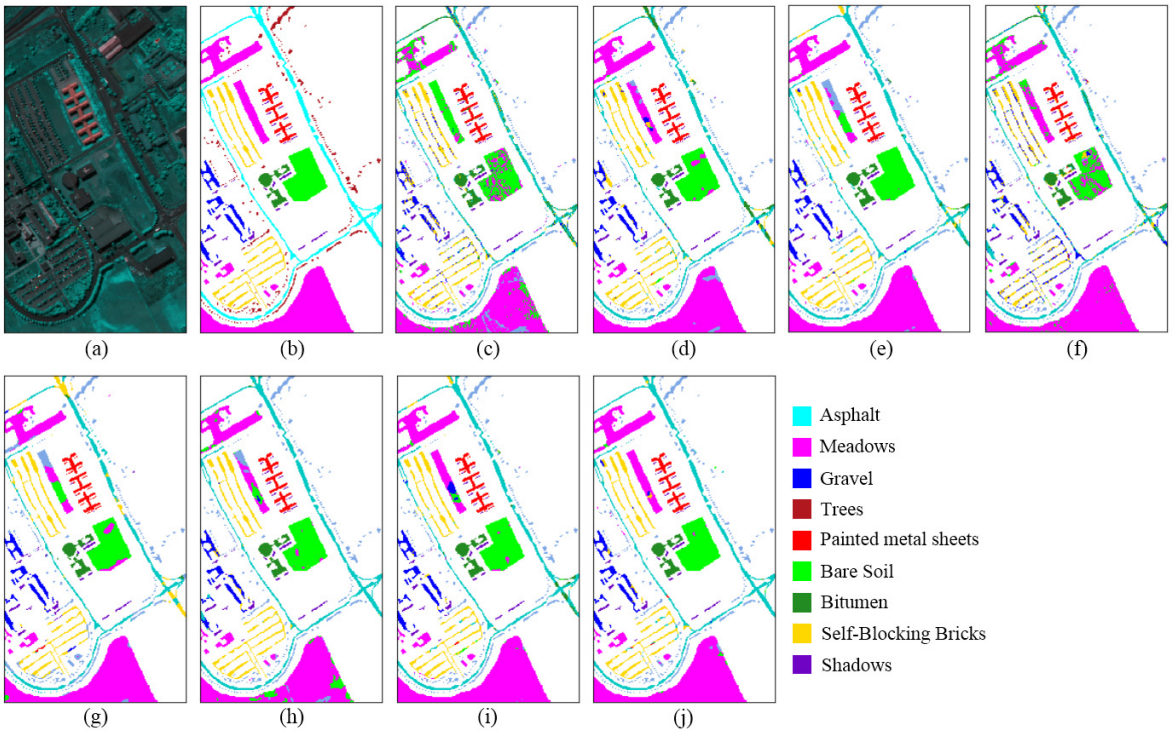


Fig. 7. Classification maps of the PU data set obtained by different methods with 20 training samples per class. (a) False-color composite image. (b) Ground truth. (c) 3-D-CNN (OA = 77.83%). (d) DCNN (OA = 92.07%). (e) DRN (OA = 93.18%). (f) CD-CNN (OA = 83.93%). (g) DFFN (OA = 87.59%). (h) SSRN (OA = 92.89%). (i) MSRN-A (OA = 95.31%). (j) MSRN-B (OA = 96.54%).

number of weight layers in the DCNN and SSRN is 10 and 12, respectively. In addition, PCA has been applied over the HSI to reduce the dimensions in the DFFN, while the 3-D-CNN, CD-CNN, and SSRN extract patches directly from the original HSI as inputs. For fair comparison, the He-initialization method is adopted in all compared methods, the batch normalization is adopted and placed after each convolutional layer in the 3-D-CNN, and the local response normalization in the CD-CNN is replaced with batch normalization. For all compared methods, we set the size of the 3-D image patches according to the corresponding references and adjust other hyperparameters (such as the learning rate and batch size) to achieve the optimal performance of each method.

The classification results of three data sets obtained by different methods are reported in Tables VI–VIII, respectively. From these tables, we can discover that the accuracies obtained by the 3-D-CNN and the CD-CNN are much lower than MSRN-A and MSRN-B for all three data sets. Because the 3-D-CNN has a much shallower architecture than the proposed MSRN, it cannot extract the multiscale features. As for the CD-CNN, all convolutional layers after the multiscale filter bank are composed of 1×1 kernels, which leads to rather limited ability to extract the spatial correlation information. For all three data sets, the MSRN-A performs better than the DRN. Because the MSRN-A can perform multiscale feature learning, it extracts more discriminative features for classification. It demonstrates the advantage of the MDSConv. In terms of averaged OA, the MSRN-A is 0.43% lower than the DFFN for the SA data set but 7.73% and 1.92% higher than that for the PU and PC data sets, respectively. In addition, the MSRN-B improves 0.18%, 8.86%, and 2.54% OA of the DFFN for the SA, PU, and PC data sets, respectively. Compared with the SSRN, both MSRN-A and MSRN-B provide better classification results for all the data sets. For example, the averaged OA of the MSRN-B is 2.76%, 3.54%, and 1.08% higher than that of the SSRN for the SA, PU, and PC data sets, respectively. Therefore, both MSRN-A and MSRN-B have better stability and adaptability than the DFFN and SSRN for HSI classification. Note that the MSRN-B surpasses the MSRN-A in performance for all data sets, which demonstrates that the HSC has a positive effect on hierarchical feature learning. All in all, for three data sets, the proposed MSRN-B has consistently achieved better performance than all compared methods.

For all three data sets, we visualize the classification results that are close to the averaged results reported in Tables VI–VIII. The corresponding classification maps of different methods are shown in Figs. 6–8. These classification maps are consistent with the corresponding results displayed in Tables VI–VIII. For instance, as can be observed from Figs. 6 and 7, both 3-D-CNN and CD-CNN cause much more noise in the classification map than other methods. In addition, compared with other methods, the MSRN-B provides cleaner classification maps and significantly reduces misclassified pixels.

Fig. 9 shows the OA of different methods with different number of training samples. All results are the average over ten repeated experiments with randomly selected training samples.

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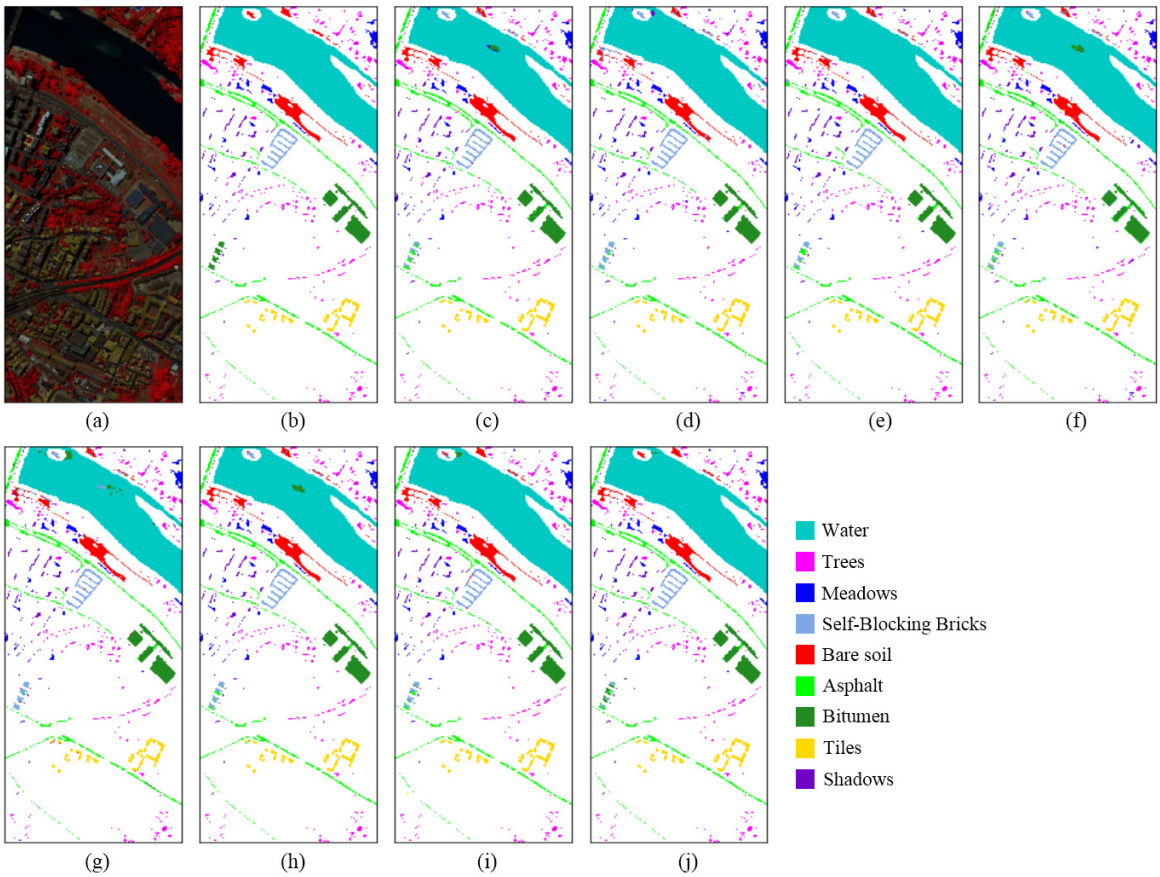


Fig. 8. Classification maps of the PC data set obtained by different methods with 20 training samples per class. (a) False-color composite image. (b) Ground truth. (c) 3-D-CNN (OA = 96.78%). (d) DCNN (OA = 96.58%). (e) DRN (OA = 97.36%). (f) CD-CNN (OA = 96.32%). (g) DFFN (OA = 96.09%). (h) SSRN (OA = 97.69%). (i) MSRN-A (OA = 98.13%). (j) MSRN-B (OA = 98.70%).

TABLE IX
TRAINING AND TESTING TIMES (IN SECONDS) OF DIFFERENT METHODS FOR THE PU DATA SET

Methods	Train Time	Test Time	OA
3D-CNN	974.25	81.55	77.85%
DCNN	307.12	186.81	92.18%
DRN	312.59	188.58	93.20%
CD-CNN	53.98	7.69	83.70%
DFFN	2319.69	1054.94	87.64%
SSRN	95191.29	1865.09	92.96%
MSRN-A	214.63	125.73	95.37%
MSRN-B	219.51	126.10	96.50%

As can be seen, as the number of training samples increases, the accuracy of all methods increases. Both MSRN-A and MSRN-B achieve high accuracies with limited training samples. For example, with ten training samples per class, the OA obtained by MSRN-A is more than 90%, 85%, and 95% for the SA, PU, and PC data sets, respectively. For each data set, the MSRN-B consistently outperforms other methods, even when the training set is very small. These results demonstrate the excellent performance of the proposed method for HSI classification.

Table IX lists the training and testing times of the proposed method and other comparison methods for the PU data set. Limited by computer memory, for each method, we divide the test samples into several batches (batch size is 256) and obtain the corresponding prediction labels. As reported in Table IX, compared with MSRN-A and MSRN-B, both DCNN and DRN need longer time for training and testing. It confirms that the MDSConv not only enhances the performance of the proposed method but also decreases the computational cost of the proposed networks. Compared with 3-D-CNN and CD-CNN, the proposed networks take longer testing time, but they significantly achieve better classification results. In addition, the proposed networks are provided with faster convergence speed than all other compared networks. The computing time of the DFFN and SSRN is much longer than other methods due to that they have a huge number of parameters. Note that MSRN-B is just a litter longer than MSRN-A in terms of the testing time, which means the HSC hardly causes extra computational cost. To sum up this section, the proposed method can greatly save computing time and provide remarkable classification performance.

E. Effect of MRB

To analyze the effect of MRB on the performance of MSRN-B, we adjust the location of MRB in MSRN-B and

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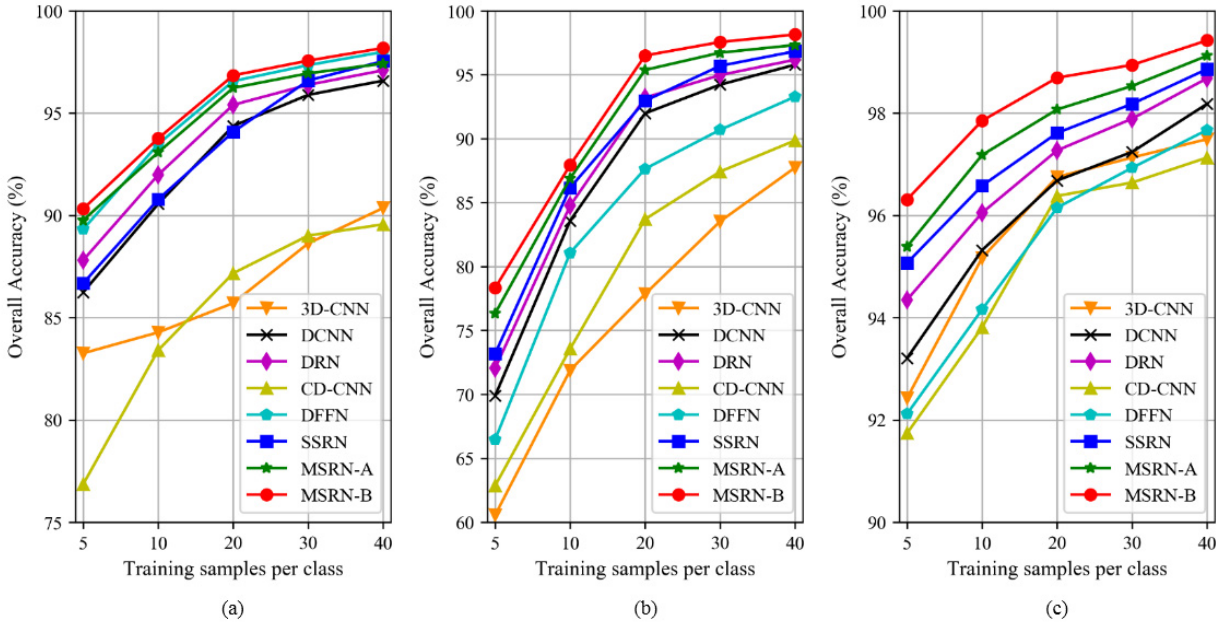


Fig. 9. OA (%) obtained by different methods with different numbers of training samples for (a) SA data sets, (b) PU data set, and (c) PC data set.

TABLE X
CLASSIFICATION RESULTS OF MSRN-B WITH DIFFERENT LOCATIONS OF MRB FOR THE PU DATA SET

Network	Location of MRB				Metrics		
	MRB-1	MRB-2	MRB-3	MRB-4	OA (%)	AA (%)	$\kappa\times100$
1	×	×	×	×	95.38±0.59	93.85±0.67	93.76±0.77
2	√	×	×	×	96.12±0.46	94.81±0.50	94.93±0.56
3	×	√	×	×	95.14±0.60	93.48±0.66	93.63±0.71
4	×	×	√	×	95.89±0.63	94.68±0.95	94.61±0.81
5	×	×	×	√	95.50±0.68	93.88±0.76	94.06±0.88
6	√	√	×	×	96.32±0.51	95.02±0.55	95.12±0.58
7	×	×	√	√	96.16±0.48	94.52±0.59	94.96±0.63
8	√	√	√	√	96.50±0.47	95.15±0.62	95.38±0.61

carry out the classification experiments on the PU data set. The corresponding results are reported in Table X. The MRB-1, MRB-2, MRB-3, and MRB-4 denote the location of the first, second, third, and fourth MRBs in MSRN-B, respectively. The $\surd$ means proposed MRB is used in the corresponding location and $\times$ means the opposite. Specifically, $\times$ means the mixed depthwise convolution in the proposed MRB is replaced with the ordinary 3×3 depthwise convolution. Therefore, network 8 corresponds to our proposed MSRN-B. All eight networks are provided with mixed depthwise convolution, except network 1. It means network 1 cannot learn multiscale features, while the other networks can do that. As shown in Table X, using MRB in different locations has different effects on the performance of the network. Networks 2 and 4 provide better results than network 1, which proves the positive effect of multiscale feature extraction. Meanwhile, it verifies the effectiveness of mixed depthwise convolution. Network 6, which carries multiscale feature learning at multiply levels of the network, performs better than networks 2 and 3. Similarly, network 7 also improves the accuracies of networks 4 and 5. Furthermore, the performance of MSRN-B (network 8) surpasses all other networks in terms of OA/AA/κ. The above-mentioned results show that, compared with only performing multiscale feature extraction once, performing it in multiple locations of the network provides greater benefits for HSI classification. In addition, those results also demonstrate the advantage of the proposed MRB.

IV. CONCLUSION

In this article, we design a novel MSRN with MDSConv for HSI classification. Specifically, we introduce DSC and design the MDSConv by mixing up the 1×1 and 3×3 filters into the depthwise convolution operation. The MDSConv can reduce the computational cost of the network and maintain promising classification accuracy. An MRB is developed by replacing the convolutional layers within an ordinary residual block with the MDSConv, which is able to learning the multiscale information from a feature map. Moreover, we

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construct the MSRN based on the MRB, which can achieve multiscale feature extraction in multiple levels of the network. In addition, the HSC is added on the cascaded two MRBs, which allows the MSRN extract more discriminative features for final classification. Experimental results on three real HSIs demonstrate the competitive advantage of the proposed method over several state-the-of-art methods in terms of both classification accuracies and computing time.

In the proposed MSRN, the 1×1 convolutional layer is used to enhance the feature representation by doubling the channel number of the feature maps. It may cause the redundancy convolution kernel, thus causing unnecessary computation cost. Therefore, we will try to find the optimal configuration of the convolution kernel number in the future work. Furthermore, we will consider introducing the attention mechanism in our future studies, so as to learn the spectral correction and spatial context information more effectively.

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